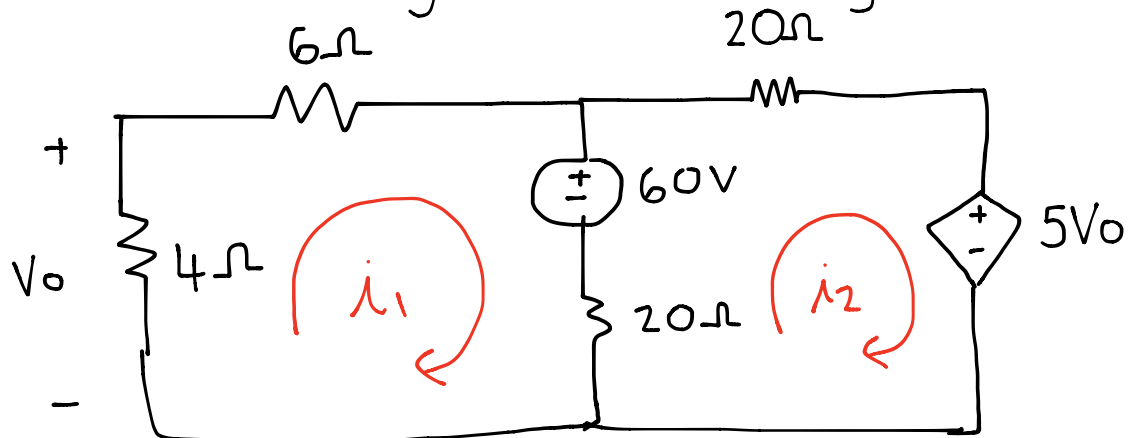


Tutorial 4 Memo

10 March 2017

3.37 Solve using Mesh Analysis



Applying mesh analysis to loop 1 and 2

$$\textcircled{1} \quad 30i_1 - 20i_2 + 60 = 0 \Rightarrow i_2 = \frac{30i_1 + 60}{20}$$

$$\textcircled{2} \quad -20i_1 + 40i_2 - 60 + 5V_o = 0$$

$$\textcircled{3} \quad * \text{ But } V_o = -4i_1 \Rightarrow i_1 = -\frac{V_o}{4}$$

Using $\textcircled{1}$ $\textcircled{2}$ and $\textcircled{3} \Rightarrow$ Solve

$$\Rightarrow \textcircled{1} \quad i_2 = \frac{3}{2} \left(-\frac{V_o}{4} \right) + 3$$

$$\Rightarrow \textcircled{2} \quad -20 \left(-\frac{V_o}{4} \right) + 40i_2 - 60 + 5V_o = 0$$

*From the new ②

$$i_2 = \frac{(-5V_0 + 60 - 5V_0)}{40}$$

$$\therefore \textcircled{2} = \textcircled{1} \quad i_2 = i_1$$

$$\frac{(-5V_0 + 60 - 5V_0)}{40} = \frac{3}{2} \left(-\frac{V_0}{4}\right) + 3 \quad \rightarrow \text{Take the 40 across}$$

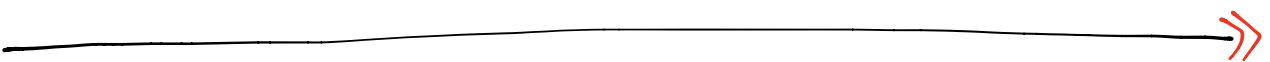
$$-5V_0 + 60 - 5V_0 = -\frac{3}{8}V_0 + 3 \quad \rightarrow \text{Get } V_0 \text{ alone}$$

$$-10V_0 + 60 = -\frac{3}{8}V_0 + 3$$

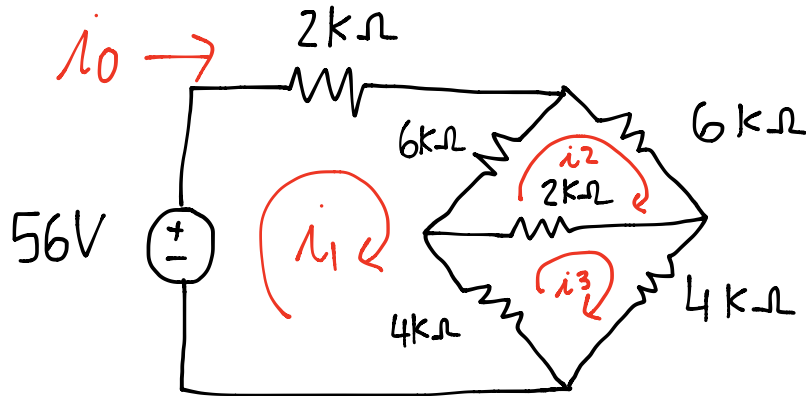
$$\left(-10 + \frac{3}{8}\right)V_0 = 3 - 60$$

$$*i_1 = \frac{-V_0}{4}$$

$$\therefore i_1 = -1.48 \text{ A} \quad V_0 = 5.922 \text{ V}$$



3.40 For the bridge network, find i_0
using mesh analysis



* Assume
all currents
in mA

→ Apply mesh analysis

$$\#1 \quad 56V = 12i_1 - 6i_2 - 4i_3 \Rightarrow 28 = 6i_1 - 3i_2 - 2i_3$$

$$\#2 \quad 0 = -6i_1 + 14i_2 - 2i_3 \Rightarrow 0 = -3i_1 + 7i_2 - i_3$$

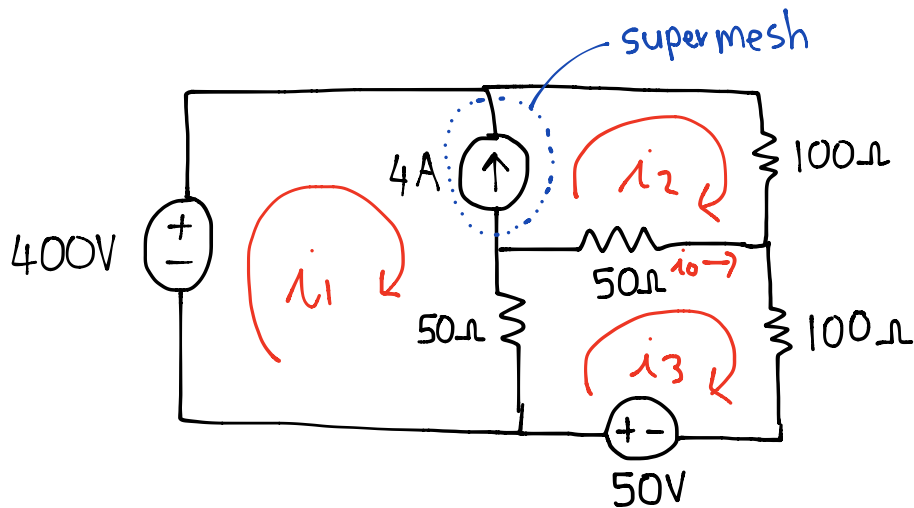
$$\#3 \quad 0 = -4i_1 - 2i_2 + 10i_3 \Rightarrow 0 = -2i_1 - i_2 + 5i_3$$

⇒ Using matrix solve

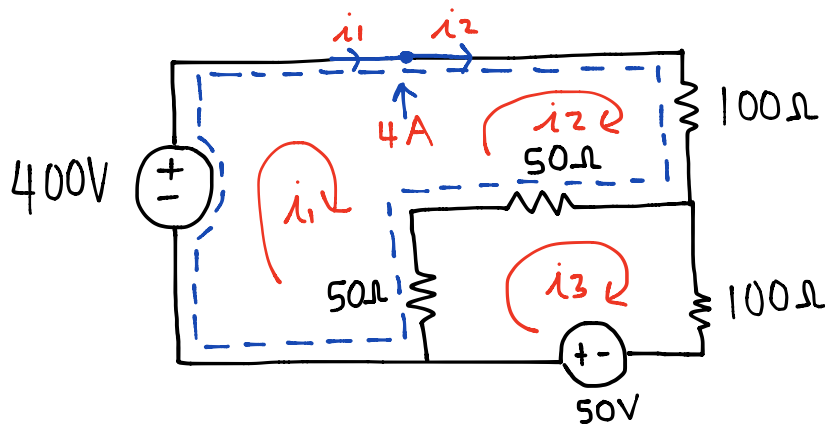
$$i_0 = i_1 = 8mA$$



3.44 Use mesh analysis to obtain $\underline{i_o}$



$\Rightarrow$ Redraw circuit for supermesh analysis



$\rightarrow$ Loop 1 and 2 form supermesh

$$-400 + 150i_2 + 50i_1 - 100i_3 = 0 \quad \dots \textcircled{1}$$

$\rightarrow$ Loop 3

$$-50 + 200i_3 - 50i_1 - 50i_2 = 0 \quad \dots \textcircled{2}$$

⇒ Also looking at the nodal current of the supermesh

$$i_1 + 4 = i_2 \dots \textcircled{3}$$

∴ solve equations

$$\textcircled{3} \text{ in } \textcircled{1} \quad -400 + 150(i_1 + 4) + 50i_1 - 100i_3 = 0 \dots \textcircled{4}$$

$$\textcircled{3} \text{ in } \textcircled{2} \quad -50 + 200i_3 - 50i_1 - 50(i_1 + 4) = 0 \dots \textcircled{5}$$

$$\textcircled{4} \quad \therefore 200 + 200i_1 - 100i_3 = 0$$

$$\textcircled{5} \quad \therefore -250 + 200i_3 - 100i_1 = 0$$

$$\therefore i_1 = \frac{-200 + 100i_3}{200} \dots \text{from } \textcircled{4}$$

$$\therefore i_1 = \frac{-250 + 200i_3}{100} \dots \text{from } \textcircled{5}$$

$$\therefore \textcircled{4} \stackrel{i_1}{=} \textcircled{5} \quad -200 + 100i_3 = -2 \times 250 + 2 \times 200i_3$$

$$i_3(100 - 400) = 200 - 500$$

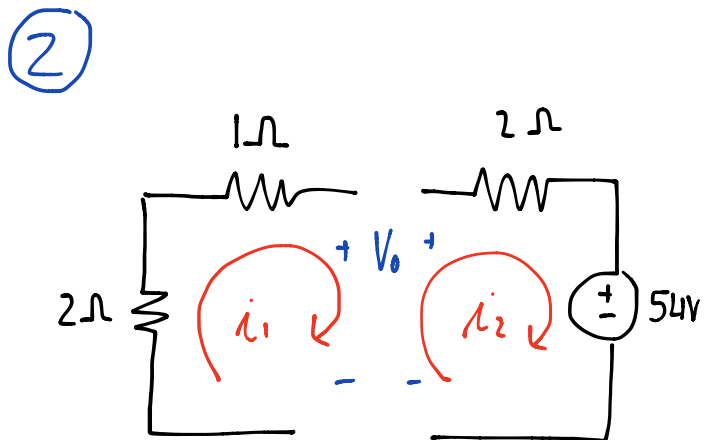
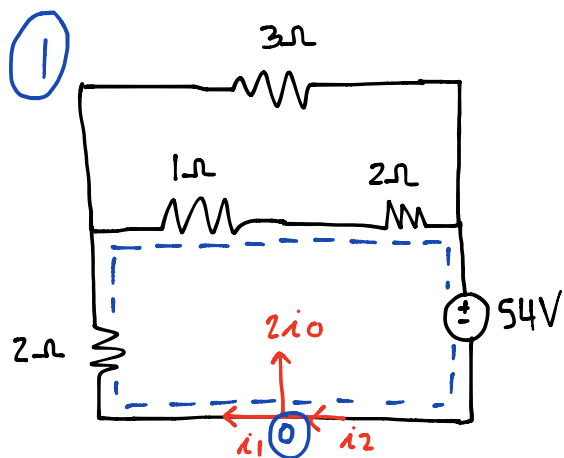
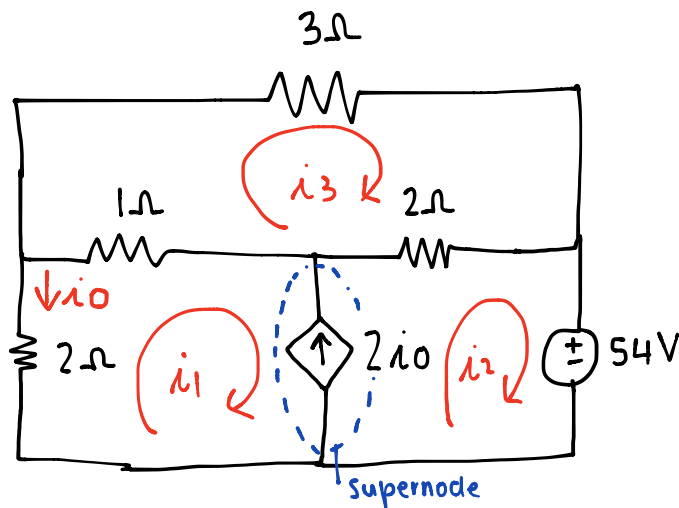
$$\therefore i_3 = 1 \text{ A} \Rightarrow i_1 = -\frac{1}{2} \text{ A}; i_2 = 3.5 \text{ A}$$

$$* i_0 = i_3 - i_2 = 1 - 3.5 = -2.5 \text{ A}$$

⇒

3.49 Find V_o and i_o

→ identify the supernode and redraw the circuit



$$* 3i_1 + 2i_2 - 3i_3 + 54 = 0 \dots \textcircled{1}$$

* node 0:

$$\left. \begin{aligned} i_2 &= i_1 + 2i_o \\ \text{and } i_1 &= -i_o \end{aligned} \right\} i_2 = -i_1 \dots \textcircled{2}$$

$$\rightarrow \text{Loop 3: } -i_1 - 2i_2 + 6i_3 = 0$$

$$\therefore 6i_3 = -i_1 \dots \textcircled{3}$$

$$i_3 = -\frac{i_1}{6}$$

→ solve equations:

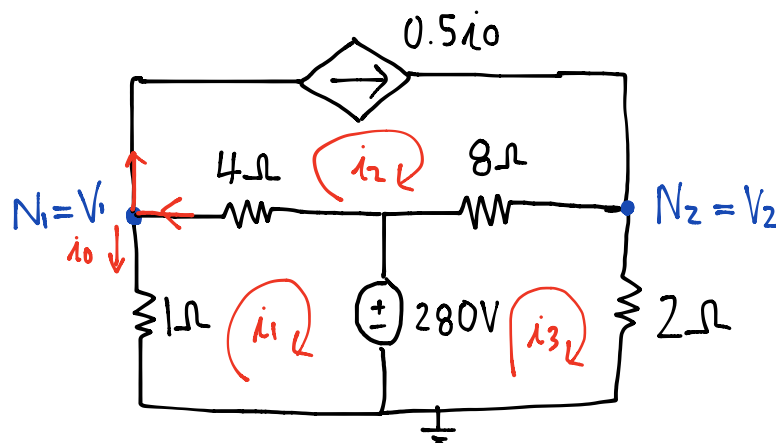
$$3(-i_2) + 2i_2 - 3\left(\frac{i_2}{6}\right) + 54 = 0$$

$$\therefore i_1 = -36$$

$$\therefore i_2 = 36$$

$$\therefore i_3 = 6$$

3.60 Calculate the power dissipated in each resistor



$$\textcircled{1} N_1: \frac{V_1}{1} + \frac{0.5i_o}{4} = \frac{280 - V_1}{4}$$

$$\begin{aligned} *NB \quad i_o = V_1/1 &\Rightarrow V_1 + 0.5V_1 = (280 - V_1)\frac{1}{4} \\ &\Rightarrow 1.75V_1 = 70 \\ &\underline{V_1 = 40V} \end{aligned}$$

$$\textcircled{2} N_2: \frac{0.5V_1}{1} + \frac{(280 - V_2)}{8} = \frac{V_2}{2}$$

$$8V_1 + 560 - 2V_2 = V_2$$

$$\therefore V_2 = \frac{8V_1 + 560}{3} = \underline{293.3V}$$

* Power dissipation

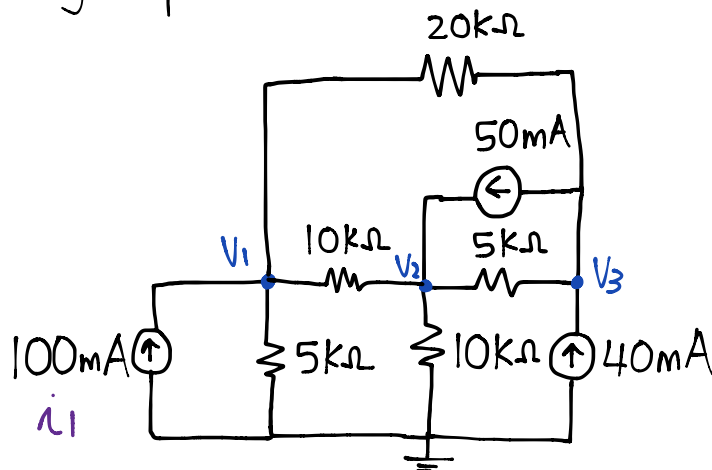
$$P_{1\Omega} = (V_1)^2/1 = 1600 \text{ Watt}$$

$$P_{2\Omega} = (V_2)^2/2 = 43 \text{ kW}$$

$$P_{4\Omega} = (280 - V_1)^2/4 = 14.4 \text{ kW}$$

$$P_{8\Omega} = (280 - V_2)^2/8 = 22.1 \text{ Watts}$$

3.69 Write the node voltage equations by inspections



* Assume all currents are in mA and voltages in V

$$* \textcircled{1} G_{11} = \frac{1}{5} + \frac{1}{10} + \frac{1}{20} = \underline{0.35}$$

$$G_{22} = \frac{1}{10} + \frac{1}{10} + \frac{1}{5} = \underline{0.4}$$

$$G_{33} = \frac{1}{5} + \frac{1}{20} = \underline{0.25}$$

* calculate all the resistances connected to a node in the reciprocal form $\frac{1}{R}$

$$* \textcircled{2} G_{12} = -\frac{1}{10} = \underline{-0.1} = G_{21}$$

$$G_{13} = -\frac{1}{20} = \underline{-0.05} = G_{31}$$

$$G_{23} = -\frac{1}{5} = \underline{-0.2} = G_{32}$$

* Calculate the resistance between nodes in $-\frac{1}{R}$

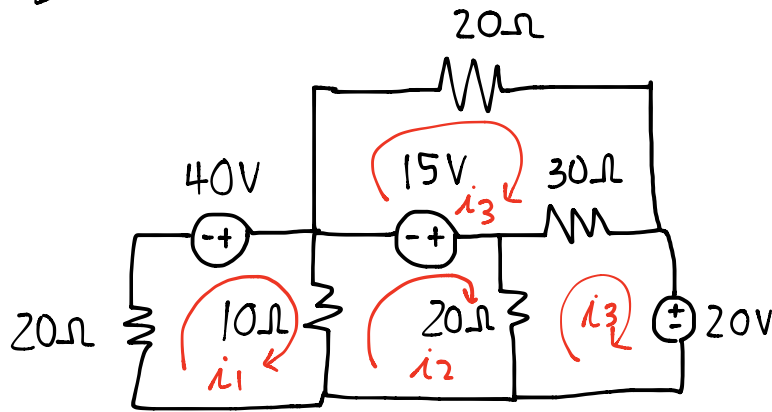
$$i_1 = 100 \quad i_2 = 50 \quad i_3 = 40 - 50 = -10$$

* node-voltage equations are :

$$\begin{bmatrix} 0.35 & -0.1 & -0.05 \\ -0.1 & 0.4 & -0.2 \\ -0.05 & -0.2 & 0.25 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 100 \\ 50 \\ -10 \end{bmatrix}$$

>>

3.72 By inspection, write mesh-current equations.



* Look at the loops:

$$\textcircled{1} R_{11} = 20 + 10 = \underline{30} \quad \textcircled{2} R_{12} = \underline{-10} = R_{21}$$

$$R_{22} = 20 + 10 = \underline{30} \quad R_{13} = \underline{0} = R_{31}$$

$$R_{33} = 30 + 20 = \underline{50} \quad R_{14} = \underline{0} = R_{41}$$

$$R_{44} = 30 + 20 = \underline{50} \quad R_{24} = \underline{0} = R_{42}$$

$$R_{23} = \underline{-20} = R_{32}$$

$$R_{34} = \underline{-30} = R_{43}$$

* $R_{ij} = R_{ji}$ for

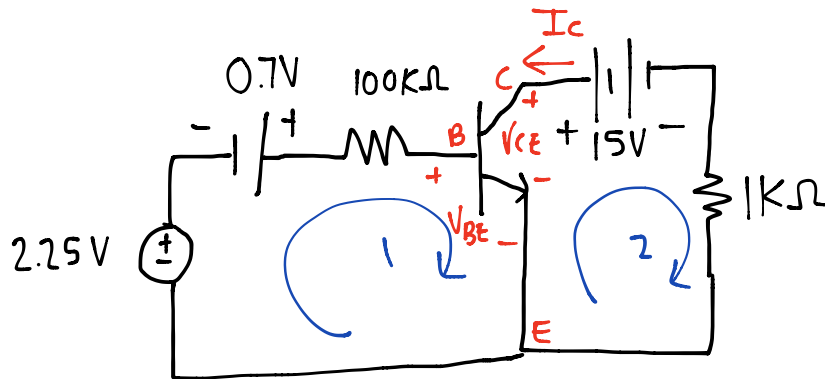
all i not equal to j

$$V_1 = 40; V_2 = 15; V_3 = -20; V_4 = -15$$

* mesh-current equations:

$$\begin{bmatrix} 30 & -10 & 0 & 0 \\ -10 & 30 & -20 & 0 \\ 0 & -20 & 50 & -30 \\ 0 & 0 & -30 & 50 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \end{bmatrix} = \begin{bmatrix} 40 \\ 15 \\ -20 \\ -15 \end{bmatrix}$$

3.89 For the transistor circuit, find I_B and V_{CE} . Let $\beta = 100$; $V_{BE} = 0.7V$



* Loop 1 KVL:

$$-2.25 - 0.7 + 100 \times 10^3 I_B + V_{BE} = 0$$

$$\rightarrow NB \quad V_{BE} = 0.7 \text{ (given)}$$

$$\therefore I_B = 22.5 \mu A$$

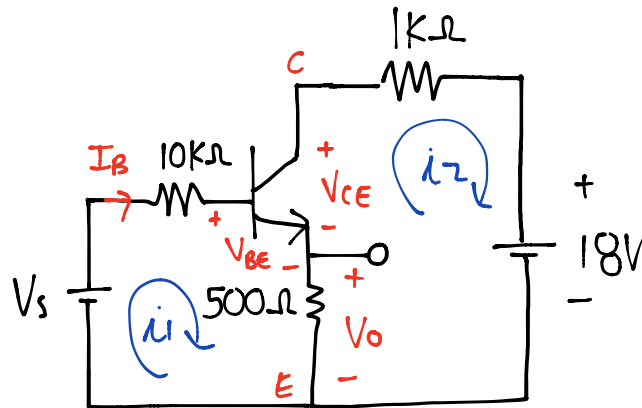
* Loop 2 KVL

$$-V_{CE} + 15 - I_C(1 \times 10^3) = 0$$

* NB $I_C = \beta I_B$ → transistor formula
 $= 100(22.5) = 2.25 \text{ mA}$

$$\therefore V_{CE} = 12.75 \text{ V}$$

3.90 Calculate V_s for the transistor
given: $v_o = 4 \text{ V}$ $\beta = 150$ $V_{BE} = 0.7 \text{ V}$



* Loop 1 KVL:

$$-V_s + 10K(I_B) + V_{BE} + I_E(500) = 0$$

* NB $I_E = (1 + \beta)I_B$

$$\therefore -V_s + 10000I_B + 0.7 + 500(1+\beta)I_B = 0$$

$$\Rightarrow V_s - 0.7 = 855,500 I_B \dots \text{eq 1}$$

$$\textcircled{2} \quad V_o = 500 I_E$$

$$= 500(151) I_B = 4$$

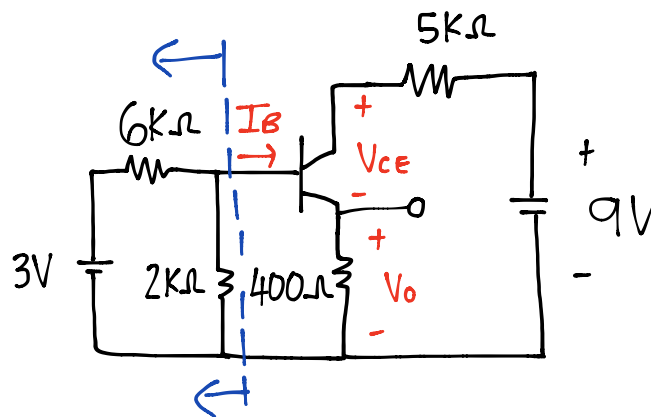
$$\therefore I_B = 5.29 \times 10^{-5}$$

$\Rightarrow \therefore$ From eq 1

$$V_s = 5.23 \text{ V}$$

3.91 For the transistor find I_B , V_{CE} , V_o

Given : $\beta = 200$, $V_{BE} = 0.7 \text{ V}$

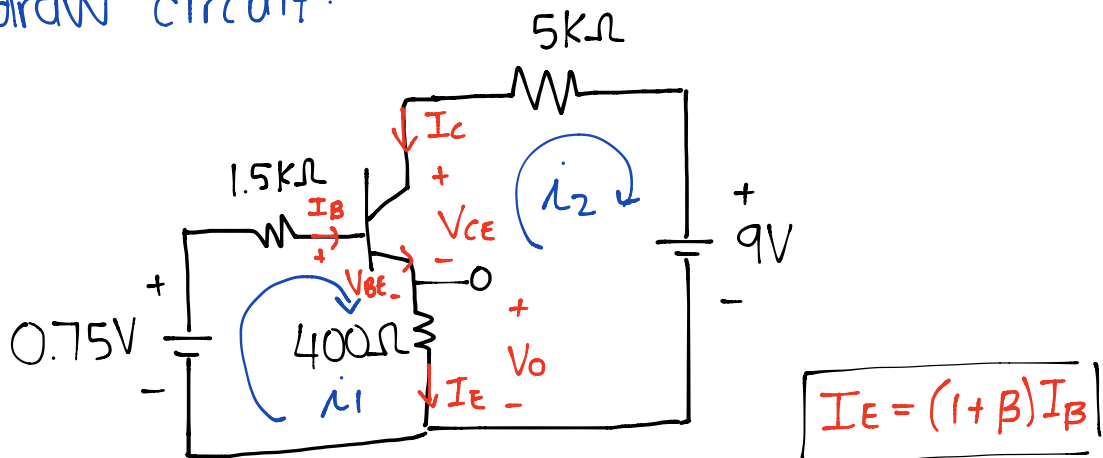


* Look at Thevenin equivalent circuit for the input

$$R_{TH} = 6 // 2 = \frac{6 \times 2}{8} = 1.5 \text{ k}\Omega$$

$$V_{TH} = \frac{2(3)}{2+6} = 0.75 \text{ V}$$

* Redraw circuit:



* KVL Loop 1:

$$-0.75 + 1.5K I_B + V_{BE} + 400 I_E = 0 \quad \text{sub}$$

$$-0.75 + 1.5K I_B + 0.7 + 400(1+\beta) I_B = 0$$

$$81900 I_B = 0.05$$

$$\therefore I_B = \underline{0.61 \mu A}$$

$$\begin{aligned} * V_O &= 400 I_E = 400(1+\beta) I_B \\ &= \underline{49 mV} \end{aligned}$$

* KVL Loop 2:

$$-400 I_E - V_{CE} - 5K I_C + 9 = 0$$

$$-400(1+\beta) I_B - V_{CE} - 5K(\beta I_B) + 9 = 0$$

$$\begin{aligned} V_{CE} &= 9 - 400(1+200) I_B - 5K(200 I_B) \\ &= 8.34 V \end{aligned}$$

$$\boxed{I_C = \beta I_B}$$